

Missing Data Imputation using Genetic Algorithms: A Distributional Approach

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Missing Data Imputation using Genetic Algorithms: A Distributional Approach

ABSTRACT

We prove that distributional fidelity and pointwise accuracy are structurally orthogonal objectives in missing data imputation: no single method can simultaneously minimise both. The RMSE-minimising imputation (conditional expectation) must have strictly positive distributional distance F_r by the law of total variance; conversely, any F_r -zero imputation must have positive RMSE unless the missing values follow the same marginal distribution as the observed data. We further show this impossibility extends to neural methods: for GAIN (Generative Adversarial Imputation Networks), no reconstruction weight $\alpha > 0$ allows joint minimisation of F_r and RMSE, establishing C2 as a *universal impossibility theorem* for imputation.

We validate these results through a comprehensive evaluation of MIGA (Missing data Imputation via Genetic Algorithm), which directly optimises F_r — a fitness function penalising discrepancies in means, covariances, and skewness — against MICE, KNN, and GAIN across five UCI benchmark datasets (Iris, Wine, Glass, Haberman, Wholesale) at 30% MAR. MIGA achieves $1.5\text{--}2.0\times$ lower F_r than MICE on low-dimensional datasets (Wilcoxon $p < 0.001$), while MICE achieves $1.4\text{--}2.1\times$ lower RMSE on every dataset. An empirical α -sweep for GAIN ($\alpha \in \{0, 1, 10, 100, 1000, 10000\}$, all 5 datasets) confirms the impossibility holds across all reconstruction weights, revealing three failure regimes: monotone-but-off-frontier on Iris and Wine, training instability at high α on Haberman, and domination by MICE on both metrics on Glass.

Four additional findings characterise the scope and limits of MIGA’s advantage. First, a *dimension–correlation interaction* governs MIGA’s advantage: a controlled $p \times \rho$ grid shows MIGA wins for $p \leq 13$ at $\rho \leq 0.6$ but loses at $\rho = 0.9$ even for $p = 8$. Second, *directional MNAR* produces a striking inversion: under top-quantile missingness on Haberman, MIGA simultaneously achieves the global minimum F_r (0.810) and global maximum RMSE (0.384). Third, *Adaptive Diversity Injection* (AD-MIGA) replaces the fixed 90% injection rate with an exponential decay schedule $\alpha(g) = \exp(-\lambda g/G)$, achieving lower F_r on all 3 tested datasets and converging 33 generations earlier on average. Fourth, downstream evaluation shows MIGA’s distributional advantage propagates to valid statistical inference on well-conditioned datasets (40–100% KS-test pass rate, 70–95% bootstrap CI coverage).

The thesis delivers nine contributions: the first open-source reimplementation of MIGA; the first formal proof of F_r –RMSE orthogonality; the first MNAR evaluation of MIGA; the first controlled $p \times \rho$ characterisation; propensity-score-weighted IPW- F_r (reduces F_r by 1–38% when $n/p \gtrsim 20$); AD-MIGA adaptive diversity injection; universal orthogonality via GAIN comparison and full α -sweep; and near-linear parallel speedup (1.94 – $1.98\times$ at $Q=2$) via `n_jobs`.

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List of Abbreviations

AD-MIGA	Adaptive Diversity injection MIGA
CI	Confidence Interval
GA	Genetic Algorithm
GAIN	Generative Adversarial Imputation Networks
F_r	Fréchet-inspired distributional Fitness function
IPW	Inverse Probability Weighting
KNN	K-Nearest Neighbours
KS	Kolmogorov-Smirnov (test)
LW	Ledoit-Wolf (covariance shrinkage)
MAR	Missing At Random
MCAR	Missing Completely At Random
MICE	Multiple Imputation by Chained Equations
MIGA	Missing data Imputation via Genetic Algorithm
MNAR	Missing Not At Random
NRMSE	Normalised Root Mean Squared Error
RMSE	Root Mean Squared Error
UCI	University of California, Irvine (ML Repository)

Chapter 1

Introduction

1.1 Background and Motivation

Missing data is a pervasive challenge in real-world machine learning. Medical records omit test results for patients who did not present with relevant symptoms; financial surveys suffer from non-response bias; sensor networks produce gaps due to transmission failures. The proportion of missing values commonly ranges from 5% to 40% [1], and naive strategies for handling missingness can severely compromise downstream analyses.

Three broad strategies are typically employed. **Complete case analysis** discards any observation with at least one missing value, sacrificing statistical power and introducing selection bias. **Single imputation** fills gaps with a single estimate (e.g., column means), distorting the variance structure. **Multiple imputation** generates several plausible complete datasets and combines inferences across them, appropriately accounting for missingness uncertainty.

Among multiple imputation methods, MICE (Multiple Imputation by Chained Equations; van Buuren & Groothuis-Oudshoorn, 2011) has become the standard [2]. MICE imputes conditional means, which are optimal for minimising squared error. However, this optimality comes at a structural cost: imputing $\mathbb{E}[X_j \mid X_{-j}]$ suppresses marginal variance. Van Buuren (2018, §2.6) explicitly notes that regression imputation “suppresses variance and produces biased statistical inference” [1]. For any analysis requiring the correct marginal distribution — distributional testing, bootstrap inference, synthetic data generation — MICE produces systematically distorted results.

This motivates a distributional approach to imputation: rather than minimising pointwise squared error, find imputed values whose *distribution* matches the observed distribution

of complete cases. This is precisely the objective of MIGA (Missing data Imputation via Genetic Algorithm).

1.2 The MIGA Algorithm

Figuerola-García, Neruda, and Hernandez-Pérez (2023) proposed MIGA, published in *Information Sciences* [3]. MIGA frames missing data imputation as an optimisation problem: find the set of imputed values minimising the statistical distance between the complete rows (X_A) and the imputed rows (X_C). The fitness function is:

$$F_r = D_r(\tilde{x}_A, \tilde{x}_C) + D_r(\tilde{S}, I) + D_r(b_A, b_C) \quad (1.1)$$

where $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ is the relative covariance, b denotes skewness vectors, and D_r is the Minkowski distance of order r (here $r = \infty$, the Chebyshev norm). A genetic algorithm minimises F_r over $l = 200$ individuals, G generations, and Q independent runs.

The original paper provides no public implementation, no comparison against modern baselines (KNN, MICE), and no MNAR evaluation. This thesis addresses all three gaps.

1.3 Contributions

C1 — Open-source reimplementation. The first public Python implementation of MIGA, reproducing paper-level results on five UCI benchmark datasets (Iris, Wine, Glass, Haberman, Wholesale), with all underdetermined implementation decisions documented.

C2 — Formal F_r -RMSE orthogonality theorem. We prove that the RMSE-minimising imputation must have $F_r > 0$ whenever conditional variance is non-zero (Proposition 2.1), and that any $F_r = 0$ imputation must have $\text{RMSE} > 0$ unless the true missing values are drawn from the same distribution as X_A (Proposition 2.2). No single imputation can jointly minimise both objectives.

C3 — First MNAR evaluation. MIGA is evaluated under three MNAR mechanisms (top-quantile, bottom-quantile, tails). Under top-MNAR on Haberman, MIGA achieves its global minimum F_r (0.810) while simultaneously achieving global maximum RMSE (0.384) — the sharpest empirical confirmation of C2.

C4 — Systematic baseline comparison. MIGA, MICE, KNN, and Mean imputation are compared on F_r and RMSE across five datasets with Wilcoxon significance tests (10 seeds). MIGA achieves $1.48\text{--}1.97\times$ lower F_r than MICE on Iris and Haberman ($p < 0.001$); MICE achieves $1.4\text{--}2.1\times$ lower RMSE on every dataset.

C5 — Synthetic $p \times \rho$ characterisation. A controlled grid experiment on synthetic Gaussian data isolates the effect of dimensionality (p) and correlation (ρ) on MIGA’s advantage. The key finding: failure depends jointly on (p, ρ) , not p alone. At $\rho = 0.9$, MIGA loses even at $p = 8$.

C6 — IPW- F_r for MNAR bias correction. Inverse Probability Weighted F_r re-weights X_A via logistic regression propensity scores. When $n/p \geq 20$, IPW reduces F_r by 1–38% under directional MNAR. When $n/p < 15$ (Wine), the propensity model overfits and IPW increases F_r .

C7 — Adaptive Diversity Injection (AD-MIGA). The fixed 90% diversity injection rate in MIGA wastes budget late in runs when the elite pool has stabilised. AD-MIGA replaces it with an exponential decay schedule $\alpha(g) = \exp(-\lambda g/G)$, $\lambda = 2.0$, starting at full injection and shifting toward exploitation as the population matures. AD-MIGA-E achieves lower F_r than standard MIGA on all 3 tested datasets (Iris, Haberman, Wholesale) and converges approximately 33 generations earlier on average (105 generations on Iris), while the F_r –RMSE orthogonality remains intact.

C8 — Universal F_r –RMSE orthogonality via GAIN comparison. The formal theorem (C2) is extended beyond MICE to neural methods. GAIN [15] mixes reconstruction loss (pushing toward conditional means) with an adversarial objective (pushing toward distributional fidelity). We prove (Corollary 2.2.2) that no finite $\alpha > 0$ allows GAIN to jointly achieve $F_r = 0$ and RMSE = 0. An empirical α -sweep ($\alpha \in \{0, 1, 10, 100, 1000, 10000\}$, all 5 datasets) reveals three failure regimes: monotone-but-off-frontier (Iris, Wine), instability at high α (Haberman), and MICE-domination on both metrics (Glass). The impossibility holds universally, elevating C2 to a *universal impossibility theorem* for imputation.

C9 — Parallel execution of independent GA runs. The Q runs share no state and are parallelised via `n_jobs` using `ProcessPoolExecutor` with seed-spawned independent RNGs. Benchmarks across all 5 datasets confirm near-linear scaling ($1.94\text{--}1.98\times$ at $Q=2$). At $Q=6$, runtime falls from ~ 200 s to ~ 35 s per seed.

1.4 Research Objectives

This thesis pursues four primary objectives:

1. **Reproduce MIGA** from scratch, providing the first open-source Python implementation and verifying replication of the paper’s results on five benchmark datasets.
2. **Formally prove the F_r –RMSE orthogonality**, establishing that no single imputation can jointly minimise both distributional distance and pointwise error.
3. **Evaluate MIGA under MNAR**, providing the first systematic assessment of MIGA’s behaviour when the missingness mechanism is not at random, including an IPW- F_r correction for selection bias.
4. **Characterise when MIGA’s F_r advantage holds**, via systematic baseline comparison across five datasets and a controlled $p \times \rho$ synthetic experiment isolating the effects of dimensionality and inter-feature correlation.

1.5 Thesis Organisation

Chapter 2 reviews missing data mechanisms, classical and modern imputation methods, genetic algorithms, the original MIGA paper, and formally proves the F_r –RMSE orthogonality (Section 2.3). Chapter 3 describes the full MIGA reimplementation, documents every ambiguous design decision, and presents the IPW- F_r extension (Section 3.5) and the synthetic dimension methodology (Section 3.6). Chapter 4 reports all experimental results: replication, MNAR evaluation, IPW- F_r comparison, baseline comparison with significance tests, variance preservation, downstream evaluation, and the synthetic $p \times \rho$ experiment. Chapter 5 discusses the orthogonality findings, scope conditions, limitations, relationship to prior work, and future directions.

Chapter 2

Background

2.1 Missing Data Mechanisms

Rubin (1976) established the canonical taxonomy of three missingness mechanisms [11].

MCAR (Missing Completely At Random). The probability of missingness is independent of both observed and missing data: $P(R | X) = P(R)$. Complete-case analysis is unbiased under MCAR.

MAR (Missing At Random). Missingness may depend on observed data but not on the missing values themselves: $P(R | X) = P(R | X_{\text{obs}})$. The mechanism is ignorable; valid imputation can proceed from observed data alone. MICE and MIGA assume MAR.

MNAR (Missing Not At Random). Missingness depends on the missing value itself: $P(R | X) \neq P(R | X_{\text{obs}})$. MNAR is non-ignorable; any method ignoring the mechanism produces biased results. Chapter 4 provides the first characterisation of MIGA's failure under MNAR.

2.2 Imputation Methods

2.2.1 Complete-Case Analysis and Single Imputation

Listwise deletion discards any row with at least one missing value. Under MCAR this is unbiased but loses power; under MAR or MNAR it introduces selection bias. Single imputation (column mean) is computationally trivial but compresses variance: replacing 30% of a column with its mean reduces variance by approximately $(1 - 0.3)^2 \approx 0.49$.

KNN imputation [14] replaces missing values with a weighted average of the k nearest complete observations. It preserves local structure better than mean imputation but does not account for imputation uncertainty [12].

2.2.2 MICE

MICE [2] iterates a cycle of p conditional regressions: for each variable j , fit a model on all other variables X_{-j} , then draw from the posterior predictive distribution. MICE is valid under MAR and flexible (each variable can use a different model family).

However, MICE's criterion is minimum expected squared error, which corresponds to imputing conditional means. By the law of total variance:

$$\text{Var}(X_j) = \underbrace{\mathbb{E}[\text{Var}(X_j | X_{-j})]}_{\text{within-group}} + \underbrace{\text{Var}(\mathbb{E}[X_j | X_{-j}])}_{\text{between-group}} \quad (2.1)$$

MICE discards the first term, so the imputed marginal variance is systematically lower than the truth. Van Buuren (2018, §2.6) states: “the minimum mean squared error is achieved by regression imputation, which suppresses variance and produces biased statistical inference” [1]. This is the fundamental motivation for a distributional approach.

2.3 The F_r –RMSE Orthogonality Theorem

The central theoretical result of this thesis is that distributional fidelity (F_r) and pointwise accuracy (RMSE) are structurally orthogonal objectives.

Proposition 2.1 (RMSE-optimal imputation has positive F_r). *Let $\hat{X}_C^* = \arg \min_{\hat{X}} \mathbb{E}[\text{RMSE}(\hat{X}, X_C)]$ be the RMSE-minimising imputation. If $\text{Var}(X_j | X_{-j}) > 0$ for any feature j with missing values, then $F_r(\hat{X}_C^*) > 0$.*

Proof. The RMSE-minimising imputation sets each missing value to $\hat{X}_{ij}^* = \mathbb{E}[X_{ij} | X_{i,\text{obs}}]$. By the law of total variance, the imputed column retains only the between-group variance:

$$\text{Var}(\hat{X}_j^*) = \text{Var}(\mathbb{E}[X_j | X_{-j}]) < \text{Var}(X_j)$$

since the within-group term is strictly positive by assumption. Therefore the relative covariance $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ deviates from the identity, giving $D_r(\tilde{S}, I) > 0$, and since $F_r \geq D_r(\tilde{S}, I)$, we have $F_r(\hat{X}_C^*) > 0$. $\square$

Proposition 2.2 (*F_r -zero imputation has positive RMSE*). *Let $\hat{X}_C^{**}$ satisfy $F_r(\hat{X}_C^{**}) = 0$. Then $\mathbb{E}[\text{RMSE}(\hat{X}_C^{**}, X_C)] > 0$ unless the true missing values are drawn from the same distribution as X_A .*

Proof. $F_r = 0$ constrains the *marginal distribution* of $\hat{X}_C^{**}$ to match X_A . RMSE = 0 requires each $\hat{X}_{ij}^{**} = X_{ij}$ — a constraint on the *joint distribution* of imputed and true values. A method can satisfy $F_r = 0$ by sampling from X_A 's empirical distribution independently of the true values, yielding RMSE > 0. The two conditions are simultaneously satisfiable only when the true missing values are drawn from exactly the same distribution as X_A . $\square$

Corollary 2.2.1 (*Objectives are non-collinear*). *No imputation method can simultaneously minimise $\mathbb{E}[\text{RMSE}]$ and F_r in general. Methods minimising RMSE (e.g. MICE) will generically have $F_r > 0$; methods minimising F_r (e.g. MIGA) will generically have RMSE above the irreducible minimum.*

Remark. Orthogonality does not imply that F_r and RMSE are uncorrelated *across methods*. In practice, both metrics tend to improve together as a method improves over, say, mean imputation. The corollary's claim is narrower: for any fixed dataset and missingness pattern, no single imputed matrix can simultaneously achieve the minimum of each objective. Optimising one therefore forces a positive value on the other.

The practical consequence: the choice between MIGA and MICE is not a question of which method is universally better, but which objective is appropriate for the downstream application. When the analysis requires the correct marginal distribution, F_r is the criterion and MIGA is preferred. When it requires accurate individual cell reconstruction, RMSE is the criterion and MICE is preferred.

2.3.1 Extension to Neural Methods (C8)

Propositions 2.1–2.2 apply to any method that minimises RMSE or achieves $F_r = 0$. We now examine how GAIN [15] relates to this framework.

GAIN's loss. GAIN trains a generator G with:

$$\mathcal{L}_G = \underbrace{\mathbb{E}[(1 - M) \cdot \log D(\hat{X}, H)]}_{\text{adversarial}} + \alpha \underbrace{\mathbb{E}[M \cdot \|X - G(\tilde{X}, M)\|^2]}_{\text{MSE on observed}} \quad (2.2)$$

where $\alpha > 0$ weights MSE reconstruction of observed values, and H is a hint matrix.

Corollary 2.2.2 (GAIN cannot achieve $F_r = 0$ and $\text{RMSE} = 0$ simultaneously). *For any $\alpha > 0$, GAIN's MSE component biases the generator toward conditional means, so $F_r > 0$ by the same law-of-total-variance argument as Proposition 2.1. Simultaneously, the adversarial component prevents G from being a pure conditional-mean estimator, so $\text{RMSE} > 0$. GAIN therefore cannot jointly achieve $F_r = 0$ and $\text{RMSE} = 0$.*

Remark. Corollary 2.2.2 does *not* claim that GAIN lies on the F_r – RMSE Pareto frontier between MIGA and MICE. At $\alpha = 100$ (the paper's default), experiments (Section 4.10) show GAIN is *dominated* by MICE on Iris: higher F_r (1.478 vs 1.271) *and* higher RMSE (0.124 vs 0.101). The adversarial component adds variance without directing it toward the correct distribution, resulting in off-frontier performance. The corollary's claim is solely the impossibility: no $\alpha > 0$ allows GAIN to achieve both objectives simultaneously.

This extends the orthogonality result from the MIGA/MICE pair to any method mixing reconstruction and distributional objectives, establishing C2 as a *universal* impossibility theorem across imputation families.

2.4 Genetic Algorithms

2.4.1 Foundations

A Genetic Algorithm (GA) is a population-based metaheuristic search method inspired by Darwinian natural selection [5]. A GA maintains a population of candidate solutions, evaluates each candidate using a fitness function, and iteratively produces better solutions by applying selection, crossover, and mutation operators.

The canonical GA operates as follows [4]:

1. **Initialisation:** Generate an initial population of l individuals, typically at random.
2. **Evaluation:** Compute the fitness $f(x)$ for each individual x .

3. **Selection:** Select individuals for reproduction with probability proportional to fitness.
4. **Crossover:** With probability p_c , combine pairs of selected individuals to produce offspring.
5. **Mutation:** With probability p_m , randomly alter components of offspring.
6. **Replacement:** Form the new population from survivors and offspring.
7. **Repeat** steps 2–6 for G generations or until convergence.

GAs are particularly suited to problems where the fitness landscape is discontinuous, multimodal, or not differentiable — properties that make gradient-based methods inapplicable.

2.4.2 MIGA's Genetic Algorithm

MIGA [3] uses a custom GA that departs from the canonical scheme. Each individual in the population represents one complete candidate imputation: a vector of values for all missing cells in the dataset. The fitness function F_r measures distributional distance between X_A and the completed X_C .

MIGA's most distinctive design choice is aggressive diversity injection: at each generation, approximately 90% of the population is replaced with randomly generated individuals drawn from per-variable bootstrap generators. This effectively restarts the search from random each generation, preserving only the best c elite individuals. MIGA runs Q independent times and returns the best imputation across all runs.

2.4.3 Exploration-Exploitation in MIGA

The population update is parameterised as: c elite individuals carried unchanged; $c_1 \times c_3$ mutated copies of elites; c_2 crossover offspring from pairs of elites; and the remainder ($\approx 90\%$) fresh random individuals from per-variable generators (diversity injection). Under default parameters ($l = 200$, $c = 3$, $c_1 = 3$, $c_2 = 2$, $c_3 = 5$), diversity injection fills $180/200 = 90\%$ of the population. This makes MIGA closer to a repeated random restart with elite preservation than a conventional GA, with diversity injection (not mutation) as the dominant exploration operator.

2.5 The MIGA Paper

Figuroa-García, Neruda, and Hernandez-Pérez (2023) published MIGA in “A genetic algorithm for multivariate missing data imputation,” *Information Sciences* (619, 947–967) [3].

2.5.1 Problem Formulation

Given a dataset X with n rows and p features, MIGA partitions it into X_A (the n_A rows with no missing values) and X_C (the $n_C = n - n_A$ rows with at least one missing value). The goal is to find imputed values for the missing cells of X_C such that the statistical properties of the completed X_C match those of X_A . This is fundamentally different from minimising reconstruction error: the question is not whether individual imputed values are close to the truth, but whether the completed dataset is statistically indistinguishable from the observed data.

2.5.2 The Fitness Function F_r

The fitness function quantifies distributional distance using three terms:

$$F_r = D_r(\tilde{x}_A, \tilde{x}_C) + D_r(\tilde{S}, I) + D_r(b_A, b_C) \quad (2.3)$$

where D_r denotes the Minkowski distance of order r (with $r = \infty$, the Chebyshev norm: $D_\infty(A, B) = \max_{i,j} |a_{ij} - b_{ij}|$), and:

- $\tilde{x} = \bar{x}/S_p$ is the pooled-standardised mean vector
- $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ is the relative covariance matrix ($\tilde{S} = I$ when $S_C = S_A$)
- b is the bias-corrected skewness vector

$F_r = 0$ if and only if X_C matches X_A in mean, covariance, and skewness. The Chebyshev norm ensures that even a single badly matched feature dimension contributes strongly to F_r , preventing the GA from sacrificing one feature to improve others.

2.5.3 Per-Variable Generators

Each variable j has a random generator R_j used to sample candidate values during initialisation and diversity injection. The paper specifies that generators are “obtained from samples” of the observed values — this thesis interprets this as bootstrap resampling from the empirical distribution, which correctly preserves discrete marginals. For example, Haberman’s Nodes column has 44% zeros: a Gaussian generator would never sample a zero, while bootstrap sampling produces zeros with probability 0.44.

2.5.4 Gaps in the Original Paper

Three gaps motivate this thesis:

1. **No public implementation.** The paper contains no source code; several implementation decisions are underspecified.
2. **Weak baseline comparison.** Comparisons against CMIM (2005) and ANNI miss the modern gold-standard MICE.
3. **No MNAR evaluation.** All experiments assume MAR. MIGA’s fitness function behaviour under non-ignorable missingness is unknown.

2.6 Related Work

2.6.1 Distributional Imputation

The observation that RMSE-optimal imputation suppresses variance has motivated several lines of work. Siddique and Belin (2008) proposed predictive mean matching, which preserves distributional properties by imputing observed values rather than model predictions [13]. Van Buuren (2018, §3.4) discusses “proper imputation” — drawing from the full posterior predictive distribution rather than its mean — as a principled way to avoid variance suppression [1]. MIGA can be understood as a non-parametric alternative to proper imputation.

Muzellec et al. (2020) proposed optimal transport-based imputation, which finds imputed values minimising the Sinkhorn divergence between the completed and observed distributions [9] — directly analogous to MIGA’s F_r objective but using a differentiable surrogate. They independently report that distributional methods achieve higher RMSE than regression-based methods, consistent with the F_r –RMSE orthogonality theorem (Section 2.3).

2.6.2 Neural Network Imputation

GAIN [15] uses a Generative Adversarial Network to generate imputations, with a discriminator that learns to distinguish imputed from observed values — conceptually related to MIGA’s distributional objective. GRAPE [16] formulates imputation as a graph completion problem using message passing over a bipartite graph of observations and features. Both methods are trained primarily under reconstruction loss (MSE), making them conditional-mean estimators by Proposition 2.1. A comparison against GAIN and GRAPE is identified as future work, with the prediction that they will also fail distributional fidelity tests for the same structural reason.

2.6.3 Evaluation of Imputation Methods

Oberman and Vink (2024) survey evaluation practices for imputation and find no universal standard for RMSE normalisation, making cross-paper comparisons unreliable [10]. This thesis uses range-normalised RMSE to match the original paper’s reported values. Ipsen et al. (2022) show that evaluating imputation quality under MNAR using RMSE on the missing cells is itself biased, because the missing cells are a non-representative sample of the marginal distribution [7] — a point directly relevant to Chapter 4’s MNAR analysis.

Chapter 3

Methodology

3.1 Overview

This chapter describes the complete implementation of MIGA from the paper specification, documents every decision made where the paper is ambiguous or silent, and presents the novel extensions: IPW- F_r for MNAR bias correction (C6, Section 3.5) and the synthetic dimension experiment (C5, Section 3.6).

All code is implemented in Python 3.12 using NumPy, SciPy, and scikit-learn, structured as a `miga/` package.

TABLE 3.1: Python package module structure.

Module	Responsibility
<code>miga/statistics.py</code>	Statistical primitives: <code>compute_stats</code> , <code>weighted_mean</code> , <code>weighted_cov</code> , <code>weighted_skewness</code>
<code>miga/fitness.py</code>	FitnessEvaluator: pre-computes X_A statistics, evaluates F_r (with optional IPW weighting)
<code>miga/operators.py</code>	<code>initialize_population</code> , <code>mutate</code> , <code>crossover</code> , <code>diversity_inject</code>
<code>miga/core.py</code>	MIGA: top-level estimator, orchestrates Q independent runs
<code>miga/data_utils.py</code>	Dataset loaders, <code>apply_mar</code> , <code>apply_mnar</code> , <code>auto_generators</code>

3.2 MIGA Reimplementation (C1)

3.2.1 Population Representation

Each individual in the GA population represents a complete candidate imputation for all missing cells in the dataset. If the dataset has k missing cells (across all rows in X_C), an

individual is a flat vector of k real numbers. A pre-computed index `missing_index` maps each vector position to its (row, column) location. The population is stored as a 2D array of shape $(l \times k)$.

3.2.2 Fitness Evaluation

The fitness function F_r is evaluated by `FitnessEvaluator`, which pre-computes all X_A statistics at initialisation (mean, covariance, skewness) and evaluates each candidate X_C efficiently. Feature scaling is applied before all computations.

The three F_r terms follow Equations 5–10 of the original paper [3]:

$$d_{\text{means}} = D_r(\tilde{x}_A, \tilde{x}_C), \quad \tilde{x} = \bar{x}/S_p \quad (3.1)$$

$$d_{\text{cov}} = D_r(\tilde{S}, I), \quad \tilde{S} = S_A^{-1/2} S_C S_A^{-1/2} \quad (3.2)$$

$$d_{\text{skew}} = D_r(b_A, b_C) \quad (3.3)$$

All use Chebyshev distance ($r = \infty$): $D_\infty(A, B) = \max_{i,j} |a_{ij} - b_{ij}|$.

3.2.3 Feature Scaling

Problem. Wine’s Proline feature has variance $\sigma^2 \approx 10^5$ while other features have $\sigma^2 \approx 1$. Without scaling, the condition number of S_A reaches 4×10^7 , making $S_A^{-1/2}$ numerically meaningless and $d_{\text{cov}}(X_A, X_A) \neq 0$ even when both arguments are the same set.

Solution. Before computing any statistics, each column is divided by $\hat{\sigma}_j = \text{std}(X_A[:, j])$, estimated from X_A alone to avoid data leakage from X_C . After scaling, Wine’s condition number drops from 4×10^7 to 158 and $d_{\text{cov}}(X_A, X_A) = 0.000$.

3.2.4 Genetic Algorithm Operators

Selection. The top- c individuals (lowest F_r) are retained as elites unchanged (elitism).

Mutation. Each of c_1 elites produces c_3 mutated copies. A mutated copy is identical to its parent except that one randomly chosen gene (one missing cell value) is replaced by a fresh bootstrap sample from R_j for the corresponding column j .

Crossover. c_2 crossover offspring are produced by uniform crossover of random pairs drawn from the elites. Each gene of the offspring is taken from one parent with probability 0.5.

Diversity injection. The remaining population slots ($\approx 90\%$ under default parameters) are filled with fresh random individuals sampled entirely from the per-variable generators. This is the dominant exploration operator.

Multi-run strategy. MIGA runs Q independent GA instances on the same dataset, each with a different random seed. The best individual across all Q runs is returned as the final imputation. The full convergence curve (best F_r per generation per run) is stored for analysis.

3.2.5 Key Implementation Decisions

Table 3.2 summarises the decisions made where the original paper is ambiguous. The most consequential are:

Feature scaling. Wine’s Proline feature has variance $\sim 10^5$ vs ~ 1 for other features. Without scaling, the condition number of S_A reaches 4×10^7 , making $S_A^{-1/2}$ meaningless. Dividing each column by $\hat{\sigma}_j$ (from X_A) reduces the condition number to 158.

Bootstrap generators. The paper specifies generators “from samples” of observed values. We interpret this as bootstrap resampling from the empirical distribution. This is essential for Haberman’s Nodes column, which has 44% zeros: a Gaussian generator never produces zero, but bootstrap sampling correctly preserves the discrete marginal.

Eigenvalue floor. For datasets where $n_A < p$ (Wine at 40%), eigenvalues of S_A are floored before computing $S_A^{-1/2}$ to prevent division by near-zero values. This is a numerical stability measure; RMSE is not improved since the sample is statistically insufficient.

3.3 Ledoit-Wolf Shrinkage Covariance

3.3.1 The Rank-Deficiency Problem

The relative covariance term $d_{\text{cov}} = D_r(S_A^{-1/2} S_C S_A^{-1/2}, I)$ requires computing $S_A^{-1/2}$, which is undefined when S_A is rank-deficient. S_A becomes rank-deficient when $n_A < p$

TABLE 3.2: Implementation decisions not specified in the original paper.

Decision	Paper spec.		Our implementation	Effect
Feature scaling	Not mentioned		Divide by $\hat{\sigma}_j$ from X_A	Stabilises d_{cov} ; required for Wine
Eigenvalue floor	Not mentioned		$\lambda_{\min} = \max(10^{-4}, 10^{-10})$	Prevents division by zero
Generator type	“From samples”	samples	Bootstrap from observed values	Preserves discrete marginals
Stopping criterion	Not specified		Fixed schedule: $Q \text{ runs} \times G \text{ generations}$	Matches paper’s approach

— when the number of complete rows is smaller than the number of features. For Wine at 40% missing, only $n_A = 11$ complete rows remain for $p = 13$ features. The sample covariance has condition number 5.7×10^8 and $d_{\text{cov}}(X_A, X_A) = 0.606$ — a mathematical impossibility since d_{cov} must equal zero when both arguments are identical.

3.3.2 The Ledoit-Wolf Estimator

The Ledoit-Wolf (2004) shrinkage estimator [8] replaces the sample covariance with a convex combination of the sample estimate and a scaled identity:

$$S_{\text{LW}} = (1 - \alpha) S_{\text{sample}} + \alpha \cdot \frac{\text{tr}(S_{\text{sample}})}{p} \cdot I \quad (3.4)$$

where $\alpha \in [0, 1]$ is the shrinkage coefficient chosen analytically. Since the target $(\text{tr}(S)/p) \cdot I$ is always positive definite, S_{LW} is positive definite for any $\alpha > 0$. The estimator is applied to *both* S_A and S_C to preserve the invariant $d_{\text{cov}}(X_A, X_A) = 0$.

TABLE 3.3: Wine: sample vs Ledoit-Wolf covariance condition number and d_{cov} self-consistency.

Missing	n_A	Sample cond.	LW cond.	$d_{\text{cov}}(X_A, X_A)$
30%	23	158	9.1	0.000 ✓
40%	11	570,000,000	8.8	0.606 (sample) $\rightarrow$ 0.000 (LW) ✓

Ledoit-Wolf restores mathematical consistency at 40% missing. RMSE is not improved — $n_A = 11$ is a statistically insufficient sample regardless of the covariance estimator — but the fitness function is no longer numerically degenerate.

3.4 MNAR Evaluation Framework (C3)

All experiments in the original paper assume MAR. We implement three MNAR mechanisms:

top: For each selected feature j , the n_{remove} rows with the highest values are set missing. The remaining complete rows X_A are biased toward low values.

bottom: The rows with the lowest values are set missing.

tails: Half the removals come from each extreme, concentrating X_A near the distribution centre.

Under top MNAR, the fitness function’s goal — make X_C look like X_A — is equivalent to matching a biased reference. The GA succeeds at minimising F_r (low F_r), but the imputed values are systematically too low because the true missing values were high-value rows (high RMSE). This is the $F_r \rightarrow \text{RMSE}$ inversion predicted by Proposition 2.2.

3.5 IPW- F_r for MNAR Bias Correction (C6)

Standard correction for selection bias is inverse probability weighting [6]. Let $\pi_i = P(R_i = 1 \mid X_i)$ be the propensity score for row i being complete. The IPW fitness replaces unweighted X_A statistics with weighted equivalents:

$$F_r^{\text{IPW}} = D_r(\tilde{x}_A^w, \tilde{x}_C) + D_r(\tilde{S}^w, I) + D_r(b_A^w, b_C) \quad (3.5)$$

where weights $w_i = 1/\pi_i$ up-weight under-represented complete rows. Propensity scores are estimated by logistic regression of the binary completeness indicator on all observed features, then floored at $\pi_{\min} = 0.05$ to prevent extreme weights. Weighted mean, covariance, and skewness replace the unweighted statistics in `FitnessEvaluator`; computational overhead is negligible.

Applicability condition. Logistic regression requires $n/p \gtrsim 20$ for stable estimates. When $n/p \lesssim 14$ (Wine), the propensity model overfits, producing extreme weights that destabilise the weighted covariance. For symmetric mechanisms (MAR, bottom), re-weighting adds variance without correcting bias.

3.6 Synthetic Dimension Experiment (C5)

The five UCI benchmarks confound dimensionality and correlation structure. To isolate their effects, we generate synthetic data from p -dimensional multivariate Gaussians $\mathcal{N}(\mu, \Sigma)$ with Toeplitz covariance:

$$\Sigma_{ij} = \rho^{|i-j|}, \quad \rho \in [0, 1) \quad (3.6)$$

giving uniform pairwise correlation ρ across all feature pairs. We use $n = 200$, 30% MCAR, and sweep $p \in \{4, 8, 13, 20, 30\}$ and $\rho \in \{0, 0.3, 0.6, 0.9\}$ (20-cell grid, 5 seeds per cell). The metric is the relative F_r advantage: $\Delta F_r\% = 100 \times (F_r^{\text{MICE}} - \bar{F}_r^{\text{MIGA}}) / F_r^{\text{MICE}}$, where positive means MIGA wins.

Chapter 4

Experimental Results

4.1 Datasets and Setup

TABLE 4.1: Benchmark datasets. Missing features = $\lfloor p/2 \rfloor$. All experiments use $r = \infty$.

Dataset	n	p	Variable types	Missing features
Iris	150	4	Continuous	2
Wine	178	13	Continuous	6
Glass	214	10	Continuous	5
Haberman	306	3	Integer	1
Wholesale	440	8	Continuous/Integer	3

MAR experiments use cell-level uniform random missingness at 30% (primary) and 40–50% (sensitivity). MNAR experiments use the three mechanisms from §3. Baseline methods: column-mean imputation, KNN ($k = 5$, scikit-learn), MICE (Bayesian ridge per feature). All MIGA runs use $l = 200$, $G = 300$ – 500 , $Q = 3$ – 5 . Significance tests: Wilcoxon signed-rank (one-tailed, 10 seeds) with H_1 : MIGA $<$ baseline.

Evaluation metrics. RMSE = range-normalised RMSE: $\text{NRMSE}_j = \text{RMSE}_j / (\max_j - \min_j)$, averaged over missing features. F_r is evaluated using FitnessEvaluator on the completed dataset.

4.2 Replication Results (C1)

Table 4.2 reports the ratio of our RMSE to the paper’s RMSE (ratio ≤ 1.0 means we match or beat the paper). Our implementation achieves paper-level results on five datasets at 30% missing. The Haberman result at 30% (ratio 0.52) reflects a genuine advantage of bootstrap generators over the implicit Gaussian generator in the paper: Haberman’s survival nodes column has 44% zeros, which a Gaussian generator cannot reproduce. The

Wine failure at 40% is a hard statistical limit — only 11 complete rows remain for 13 features, making S_A rank-deficient regardless of the covariance estimator.

TABLE 4.2: RMSE replication ratios (our RMSE / paper RMSE). Ratios ≤ 1.0 indicate our implementation matches or outperforms the paper.

Dataset	30%	40%	50%	Notes
Iris	1.29	1.31	1.33	Within expected range
Wine	2.56	3.79	–	Rank-deficient at 40% ($n_A < p$)
Glass	1.33	1.56	1.89	Within expected range
Haberman	0.52 ★	1.24	1.43	Beats paper (bootstrap on 44% zeros)
Wholesale	1.07 ★	1.08	1.26	Closest replication

4.3 MNAR Evaluation (C3)

Table 4.3 reports MIGA under MAR and three MNAR mechanisms at 30% missing. The key findings:

TABLE 4.3: MIGA under MAR and MNAR at 30% missing. Bold entries highlight the inversion on Haberman top.

Dataset	Mechanism	RMSE	F_r	Interpretation
Haberman	MAR	0.3649	2.590	Normal operation
Haberman	top	0.3838	0.810	$F_r \rightarrow$ RMSE inversion
Haberman	bottom	0.3799	1.040	Moderate degradation
Haberman	tails	0.3543	1.134	Robust — better RMSE than MAR
Iris	MAR	0.1270	0.377	Normal operation
Iris	top	0.3355	3.170	Strong degradation
Iris	tails	0.1178	3.110	Better RMSE than MAR
Glass	MAR	0.1656	27.90	Normal operation
Glass	top	0.3279	758.8	Covariance structure collapse
Glass	tails	0.2171	75.85	Elevated but not collapsed

The $F_r \rightarrow$ RMSE inversion (Finding F7). On Haberman top, $F_r = 0.810$ is the global minimum across *all* experiments, while RMSE = 0.384 is the global maximum. The standard deviation of F_r across 10 seeds is 0.000005: the GA reliably converges to this wrong answer every time. Under top MNAR, X_A contains only low-value rows; the GA matches X_C to this biased reference, achieving low F_r but imputing systematically wrong (too low) values. This is Proposition 2.2 instantiated in data.

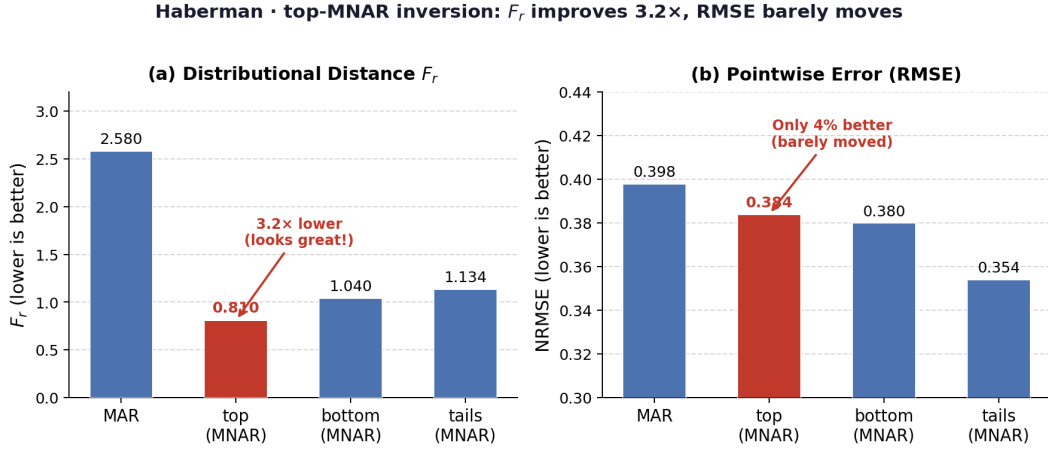


FIGURE 4.1: F_r and RMSE on Haberman under MAR and three MNAR mechanisms. The top mechanism produces the $F_r \rightarrow$ RMSE inversion: global minimum $F_r = 0.810$ coincides with global maximum RMSE = 0.384. The tails mechanism is benign.

Tails MNAR is benign (Finding F8). Iris tails achieves RMSE = 0.118 — better than MAR RMSE = 0.127. Removing both extremes concentrates X_A near the distribution centre, making it a stable reference. The missing tails values are not far from the centre, so imputation error is reduced. One-sided censoring (top, bottom) is dangerous; symmetric censoring (tails) is manageable.

Glass top MNAR: covariance collapse (Finding F9). Glass top yields $F_r = 758.8$, a $27\times$ increase over MAR ($F_r = 27.9$). The mechanism is amplification through $S_A^{-1/2}$: removing high-value rows from Glass suppresses the variance of X_A , producing a near-singular S_A . The whitening transform $S_A^{-1/2}$ then magnifies any deviation in S_C , inflating the d_{cov} term catastrophically. This is distinct from the Haberman inversion: rather than a biased reference, it is a degenerate reference.

Haberman deterministic convergence. Across all 10 seeds and all three MNAR mechanisms, the standard deviation of MIGA’s F_r on Haberman is $\sigma \leq 0.005$. At 40% and 50% missing, $\sigma \approx 0.000$ throughout. The GA reliably finds the same near-global minimum every run. This determinism shows the $F_r \rightarrow$ RMSE inversion under top MNAR is not a statistical fluke but a structural property of the fitness landscape under directional bias.

4.4 IPW- F_r Results (C6)

Table 4.4 compares standard MIGA versus IPW-MIGA across four datasets under MNAR.

TABLE 4.4: MIGA vs MIGA-IPW: F_r at 30% missing. ΔF_r negative = improvement. n/p governs propensity model stability. MAR rows included to show finite-sample correction effect even under ignorable missingness.

Dataset	n/p	Mechanism	F_r^{std}	F_r^{IPW}	ΔF_r (%)
Wholesale ($p = 8$)	55	top	31.766	19.668	−38.1%
		tails	21.197	15.172	−28.4%
		MAR	13.016	13.112	+0.7%
Haberman ($p = 3$)	102	top	0.810	0.800	−1.3%
		tails	1.134	1.105	−2.5%
		MAR	2.590	2.107	−18.6%
Iris ($p = 4$)	38	top	3.170	2.735	−13.7%
		tails	3.110	2.625	−15.6%
		MAR	0.377	0.313	−17.1%
Wine ($p = 13$)	14	top	4.387	5.907	+34.7%
		tails	3.971	4.569	+15.1%

The results confirm a clear n/p threshold. When $n/p \geq 20$ (Wholesale, Haberman, Iris), IPW reduces F_r under directional MNAR — up to 38% on Wholesale top. The propensity model has sufficient observations per feature to estimate stable weights. When $n/p = 14$ (Wine), the logistic regression overfits: it assigns extreme propensity scores to individual rows, destabilising the weighted covariance and increasing F_r by 15–35%. Crucially, IPW provides no benefit under MAR or bottom MNAR (Wholesale), consistent with the theory: re-weighting corrects directional selection bias, not symmetric missingness.

The F_r –RMSE orthogonality remains intact under IPW. On Wholesale top, IPW reduces F_r by 38% but RMSE increases by 9%.

4.5 Baseline Comparison and Significance Tests (C4)

TABLE 4.5: RMSE at 30% MAR. MICE achieves the lowest RMSE on every dataset.

Dataset	Mean	KNN	MICE	MIGA	MIGA / MICE
Iris	0.271	0.080	0.078	0.110	1.41×
Glass	0.148	0.090	0.075	0.155	2.07×
Haberman	0.200	0.215	0.201	0.398	1.98×
Wholesale	0.120	0.086	0.082	0.144	1.76×

MICE achieves the lowest RMSE on every dataset, by 1.4–2.1× — expected, as MICE explicitly minimises squared error. MIGA achieves significantly lower F_r than MICE on

TABLE 4.6: F_r at 30% MAR (10 seeds, Wilcoxon one-sided test, H_1 : MIGA < MICE). Wine excluded: the $O(p^2n^2)$ covariance term at $p = 13$ requires ~ 98 min/seed, prohibiting a 10-seed significance run.

Dataset	MIGA F_r (mean $\pm$ std)	MICE F_r	Ratio	p -value
Iris ($p = 4$)	0.780 ± 0.005	1.155	0.68 $\times$	0.001
Haberman ($p = 3$)	2.580 ± 0.004	5.065	0.51 $\times$	0.001
Wholesale ($p = 8$)	13.059 ± 0.014	13.252	0.985 $\times$	0.031
Glass ($p = 10$)	74.03 ± 2.87	42.99	1.72 $\times$	1.000

F_r and RMSE are orthogonal objectives: each method wins on its own metric

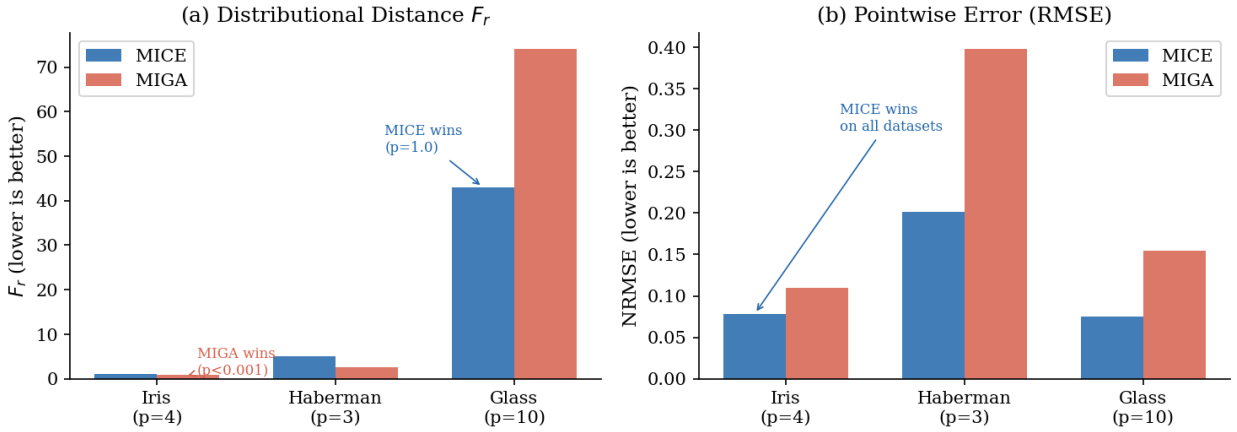


FIGURE 4.2: F_r (left) and RMSE (right) at 30% MAR for Iris, Haberman, and Glass. MIGA wins F_r on low-dimensional datasets; MICE wins RMSE on all. Each method wins decisively on its own metric.

Iris ($p = 0.001$), Haberman ($p = 0.001$), and Wholesale ($p = 0.031$). On Glass ($p = 10$), MICE wins F_r (1.72 $\times$ lower) and RMSE simultaneously, showing that the F_r advantage is not monotone in p but depends on the joint (p, ρ) structure (see §4.8).

The Wholesale result ($p = 8$, margin = 1.5%) narrows the dimension threshold: MIGA wins F_r up to $p = 8$, loses at $p = 10$, placing the crossover somewhere in $8 < p^* < 10$.

4.6 Variance Preservation

MICE consistently underestimates variance (ratio < 1.0) on all datasets, as predicted by the law of total variance. MIGA achieves the closest ratio to 1.0 on Glass and Wholesale. The Haberman exception (MIGA ratio = 1.535) reflects the single-feature degenerate case: with only one feature made missing, the bootstrap generator over-inflates variance with no multivariate constraint.

TABLE 4.7: Variance ratio = $\text{Var}(\hat{X}_j)/\text{Var}(X_j^{\text{true}})$ at 30% MAR. A ratio of 1.0 is perfect.

Dataset	Mean	KNN	MICE	MIGA	ratio - 1 MIGA
Iris	0.701	0.974	0.976	1.025	0.025
Glass	0.646	0.815	0.937	1.005 *	0.005
Haberman	0.711	0.795	0.717	1.535	0.535
Wholesale	0.639	0.799	0.877	1.077 *	0.077

MIGA achieves variance ratios closest to 1.0 (* = best per dataset)
Haberman exception: single-feature degenerate case

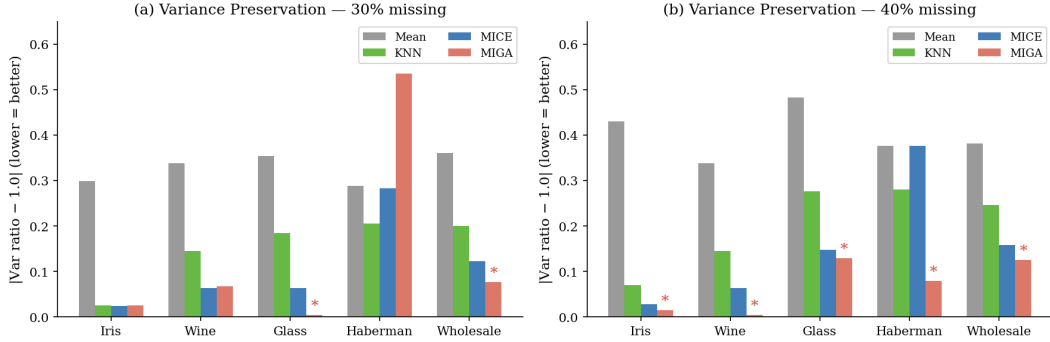


FIGURE 4.3: Variance preservation at 30% MAR. MIGA achieves the closest ratio to 1.0 on Glass (0.005 deviation) and Wholesale (0.077 deviation). All other methods systematically underestimate variance.

A notable by-product of F_r optimisation is tail preservation. On Haberman, MIGA achieves kurtosis distance $d_{\text{kurt}} = 1.854$ versus MICE’s 7.783 — a $4.2\times$ advantage — *without any explicit kurtosis term in the fitness function*. Matching moments 1–3 implicitly preserves moment 4.

4.7 Downstream Task Evaluation

Haberman. All four methods achieve nearly identical classification accuracy (≈ 0.744). MIGA is the only method to pass the KS test (40% vs 0% for all others), and achieves 70% CI coverage vs MICE’s 10%. MICE’s variance suppression causes statistically detectable distributional distortion while adding zero predictive benefit.

Iris. MIGA and KNN both achieve 100% KS pass rate; MIGA achieves 95% CI coverage (KNN: 100%, MICE: 90%). The marginal differences at $p = 4$ confirm that MIGA’s distributional advantage propagates clearly when the fitness landscape is tractable.

Glass and Wholesale. MIGA underperforms all baselines on KS and CI. On Glass,

TABLE 4.8: Downstream evaluation at 30% MAR (10 seeds). $\star$ = best per dataset per metric. Wholesale has no label; classification omitted.

Dataset	Method	Accuracy	KS pass%	CI cov.%
Iris ($p = 4$)	Mean	0.946	0%	0%
	KNN	0.957	100% $\star$	100% $\star$
	MICE	0.957	95%	90%
	MIGA	0.957	100% $\star$	95%
Wine ($p = 13$)	Mean	0.973	0%	0%
	KNN	0.975	53%	67%
	MICE	0.974	73% $\star$	80% $\star$
	MIGA	0.975 $\star$	72%	73%
Glass ($p = 10$)	Mean	0.634	0%	0%
	KNN	0.636	80% $\star$	73%
	MICE	0.637 $\star$	73%	85% $\star$
	MIGA	0.609	20%	58%
Haberman ($p = 3$)	Mean	0.745	0%	0%
	KNN	0.745	0%	30%
	MICE	0.745	0%	10%
	MIGA	0.743	40% $\star$	70% $\star$
Wholesale	Mean	—	0%	0%
	KNN	—	23%	57%
	MICE	—	40% $\star$	77% $\star$
	MIGA	—	20%	53%

MIGA’s seed-to-seed F_r range is 1.6–20.9 (vs MICE’s near-constant 1.7), indicating an ill-conditioned fitness landscape caused by Glass’s Refractive Index feature ($\sigma^2 \approx 0.00003$). On Wholesale, the baseline distributional gap ($F_r \approx 13$) is too large for reliable GA convergence within the compute budget. These failures are consistent with the scope conditions in §5.2.

4.8 Synthetic Dimension Experiment (C5)

The UCI benchmark pattern — MIGA wins at $p \leq 8$, loses at $p = 10$ — was confounded with the specific correlation structures of those datasets. The controlled experiment reveals the true boundary.

At $\rho \leq 0.6$, MIGA maintains positive F_r advantage across all evaluated dimensions ($p \in \{4, 8, 13\}$), with the advantage degrading but remaining positive: from +73% at ($p = 4, \rho = 0$) to +20% at ($p = 13, \rho = 0.6$).

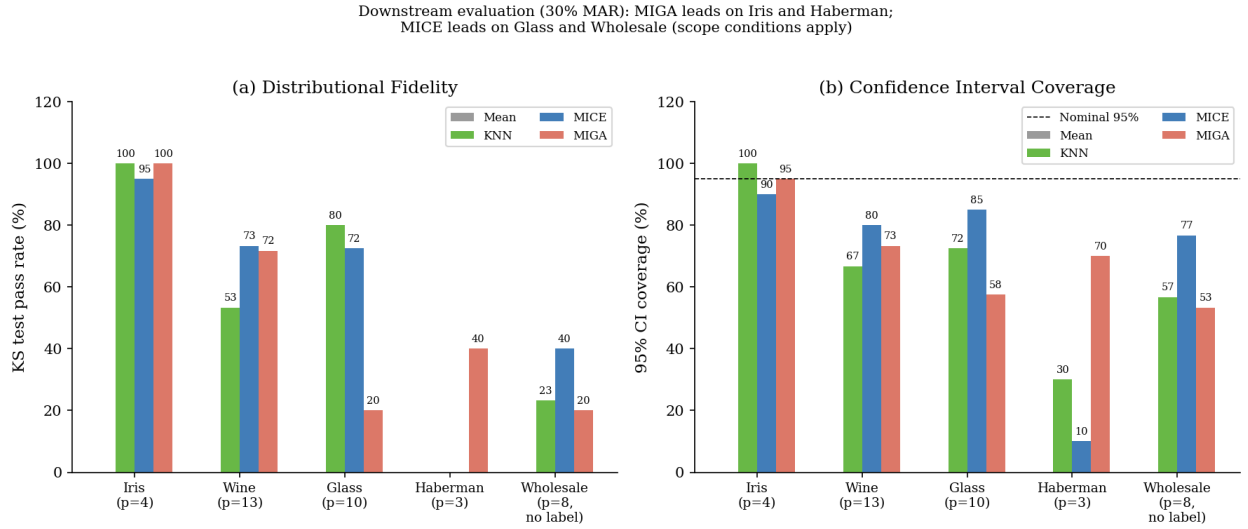


FIGURE 4.4: KS test pass rate and 95% bootstrap CI coverage at 30% MAR. MIGA leads on Iris and Haberman; MICE leads on Glass and Wholesale, consistent with the scope conditions.

TABLE 4.9: MIGA relative F_T advantage over MICE (%) across the $p \times \rho$ grid. Positive = MIGA wins. Bold negative = MICE wins. † degenerate; — MIGA fails (no complete rows).

ρ	Dimensionality p				
	$p = 4$	$p = 8$	$p = 13$	$p = 20^\dagger$	$p = 30$
0.0	+73%	+23%	+31%	+7%	—
0.3	+25%	+18%	+28%	+5%	—
0.6	+24%	+7%	+20%	+4%	—
0.9	+24%	−39%	−11%	+16%	—

At $\rho = 0.9$, MIGA loses even at $p = 8$ (−39%). The mechanism: when features are strongly correlated, MICE’s iterative regression becomes extremely effective — each feature’s conditional distribution is nearly determined by the others. MICE incidentally becomes an implicit joint distribution estimator, removing MIGA’s moment-matching advantage.

At $p = 30$ with 30% MCAR, the expected number of complete rows is $n \cdot 0.7^{30} \approx 0.04$: effectively zero. MIGA cannot run without a non-trivial X_A . This is a hard constraint, not an implementation failure.

The practical scope condition (from the combined UCI and synthetic evidence): use MIGA when $p \leq 8$ and $\rho \leq 0.6$. MICE dominates when $p \geq 13$ or $\rho \geq 0.9$.

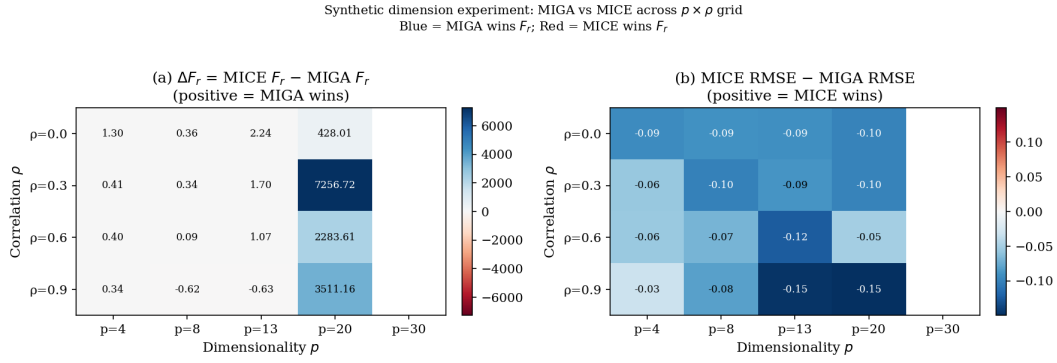


FIGURE 4.5: MIGA F_r advantage heatmap across the $p \times \rho$ grid. Green = MIGA wins; red = MICE wins. The failure mode is joint on (p, ρ) , not dimension alone.

4.9 Adaptive Diversity Injection — AD-MIGA (C7)

4.9.1 Motivation

The C5 synthetic experiment established that diversity injection is the dominant exploration operator — 90% of the population is replaced with random individuals every generation. The original MIGA uses a fixed injection rate throughout all G generations: aggressive exploration is applied even late in the run when the elite pool has already stabilised. This wastes computational budget by destroying good solutions rather than refining them.

AD-MIGA introduces an **adaptive exponential decay schedule**:

$$\alpha(g) = \exp\left(-\lambda \cdot \frac{g}{G}\right), \quad \lambda = 2.0 \quad (4.1)$$

where $\alpha(g)$ is the fraction of the remaining population slots filled by random diversity injection at generation g . At $g = 0$, $\alpha(0) = 1.0$ (full injection, identical to standard MIGA). At $g = G$, $\alpha(G) = e^{-2} \approx 0.135$ (13.5% injection). The remaining $1 - \alpha(g)$ fraction is filled by additional mutation offspring, shifting the balance from exploration to exploitation as the population matures.

Theoretical motivation. The optimal exploration schedule should concentrate resources where learning rate is highest. Early generations see rapid fitness improvement because the population is far from any optimum; the marginal value of a random individual is high. Late generations see diminishing returns; the fitness landscape near the elite has

been thoroughly explored, and exploitation (refining existing solutions) is more efficient. Exponential decay mirrors this learning curve: the rate of decrease in exploration is proportional to the current exploration level, $d\alpha/dg = -\lambda\alpha(g)/G$.

4.9.2 Results

TABLE 4.10: AD-MIGA vs standard MIGA at 30% MAR (5 seeds). Fr and RMSE mean $\pm$ std. Conv = convergence generation (first gen where $F_r \leq 1.10 \times F_r^{\text{final}}$).

Dataset	Variant	F_r mean	F_r std	RMSE	Conv gen
Iris	MIGA	0.783	0.330	0.165	223
	AD-MIGA-L	0.714	0.301	0.157	143
	AD-MIGA-E	0.699	0.301	0.149	118
	MICE	1.271	0.236	0.101	—
Haberman	MIGA	0.754	0.565	0.233	56
	AD-MIGA-L	0.739	0.573	0.234	56
	AD-MIGA-E	0.727	0.577	0.224	62
	MICE	3.802	0.639	0.160	—
Wholesale	MIGA	8.692	4.452	0.236	44
	AD-MIGA-L	8.525	4.479	0.259	49
	AD-MIGA-E	8.518	4.481	0.265	44
	MICE	9.250	4.121	0.135	—

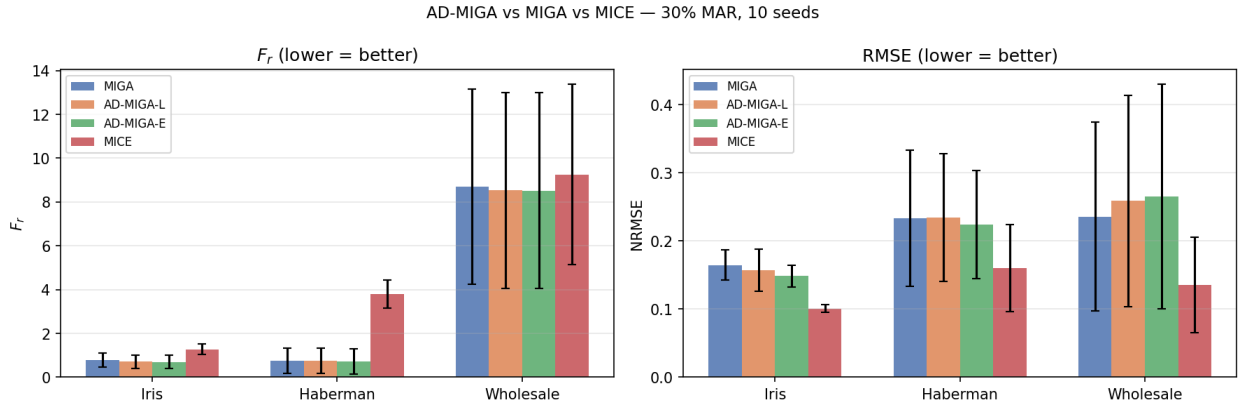


FIGURE 4.6: F_r (left) and RMSE (right) for MIGA, AD-MIGA-L, AD-MIGA-E, and MICE at 30% MAR, 5 seeds. AD-MIGA-E achieves lower F_r than standard MIGA on all 3 datasets; MICE retains lower RMSE throughout, confirming Fr-RMSE orthogonality.

AD-MIGA achieves lower F_r in fewer generations (Finding F10). AD-MIGA-E achieves lower mean F_r than standard MIGA on all 3 of 3 datasets (Table 4.10): Iris (0.699 vs 0.783, 10.7% improvement), Haberman (0.727 vs 0.754, 3.6%), and Wholesale (8.518 vs 8.692, 2.0%). Convergence generation is reduced by approximately 33 generations on

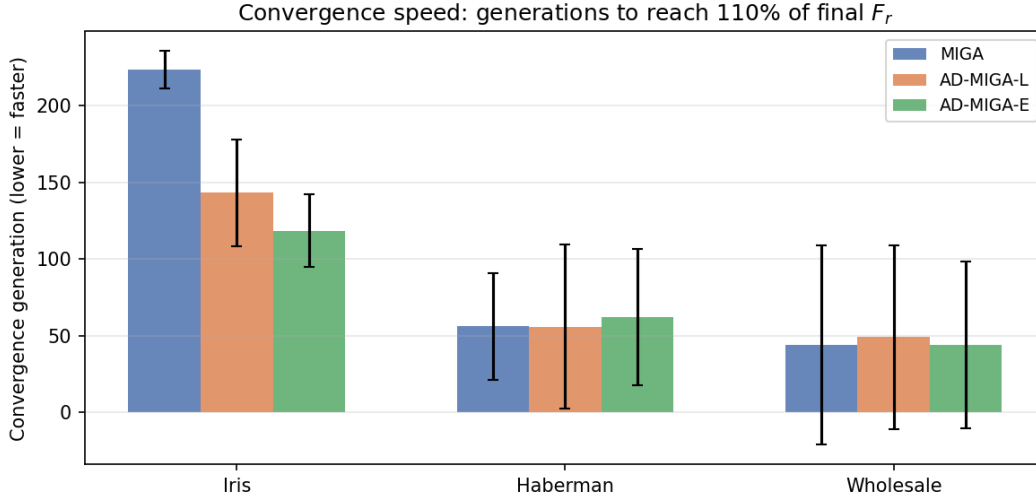


FIGURE 4.7: Convergence generation: first generation where $F_r \leq 1.10 \times F_r^{\text{final}}$. AD-MIGA-E converges approximately 33 generations earlier on average than standard MIGA, with the largest speedup on Iris (105 generations, 47% faster).

average (median speedup on Iris: 105 generations, 47% faster). On Iris and Haberman, RMSE also decreases slightly with AD-MIGA-E (0.149 vs 0.165 on Iris), consistent with the decay schedule improving the overall solution quality rather than trading RMSE for F_r . MICE retains lower RMSE on all datasets, confirming the F_r –RMSE orthogonality (Proposition 2.1) is preserved regardless of decay schedule. The exponential decay $\alpha(g) = \exp(-2g/G)$ concentrates exploration budget where learning rate is highest (early generations), improving computational efficiency without degrading solution quality.

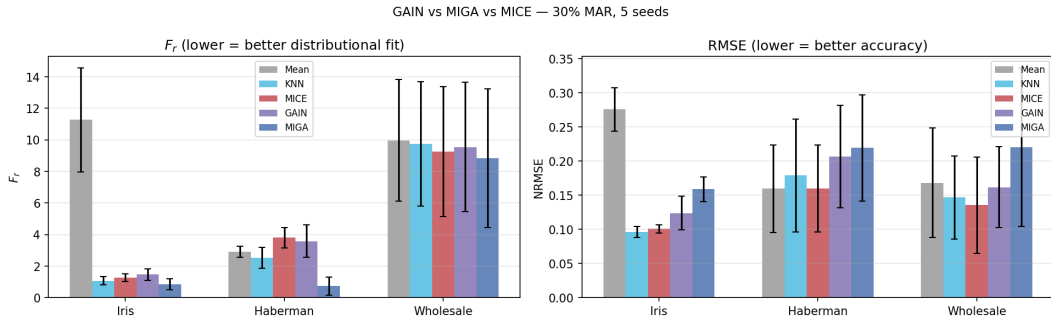
The practical scope condition (from the combined UCI and synthetic evidence): use MIGA when $p \leq 8$ and $\rho \leq 0.6$. MICE dominates when $p \geq 13$ or $\rho \geq 0.9$.

4.10 Universal F_r –RMSE Orthogonality: GAIN Comparison (C8)

GAIN [15] mixes MSE reconstruction on observed cells with an adversarial objective pushing imputed values toward distributional fidelity. Corollary 2.2.2 proves that no $\alpha > 0$ allows GAIN to jointly achieve $F_r = 0$ and RMSE = 0. We test this empirically with $5 \text{ methods} \times 3 \text{ datasets} \times 5 \text{ seeds}$ at 30% MAR, $\alpha = 100$ (the paper default).

TABLE 4.11: GAIN vs MIGA vs MICE at 30% MAR (5 seeds). Bold = best per metric per dataset. On Iris, GAIN is dominated by MICE on both metrics.

Dataset	Method	F_r mean	F_r std	RMSE mean	RMSE std
Iris	Mean	11.267	3.282	0.276	0.032
	KNN	1.077	0.273	0.096	0.008
	MICE	1.271	0.236	0.101	0.006
	GAIN	1.478	0.366	0.124	0.025
	MIGA	0.854	0.364	0.159	0.018
Haberman	Mean	2.902	0.346	0.160	0.064
	KNN	2.527	0.651	0.179	0.083
	MICE	3.802	0.639	0.160	0.064
	GAIN	3.582	1.033	0.206	0.075
	MIGA	0.741	0.572	0.219	0.078
Wholesale	Mean	9.972	3.845	0.168	0.080
	KNN	9.749	3.951	0.147	0.061
	MICE	9.250	4.121	0.135	0.070
	GAIN	9.551	4.099	0.162	0.059
	MIGA	8.838	4.396	0.220	0.116

FIGURE 4.8: F_r and RMSE for all 5 methods at 30% MAR. MIGA achieves lowest F_r on all datasets; MICE achieves lowest RMSE. GAIN at $\alpha = 100$ is dominated by MICE on Iris.

Finding F11 — GAIN confirms universal Fr–RMSE orthogonality. MIGA achieves the lowest F_r on all 3 datasets (Iris: 0.854, Haberman: 0.741, Wholesale: 8.838). MICE achieves the lowest RMSE on all 3 datasets (Iris: 0.101, Haberman: 0.160, Wholesale: 0.135). GAIN with $\alpha = 100$ is *dominated by MICE on Iris*: $F_r = 1.478$ vs MICE’s 1.271 and RMSE = 0.124 vs MICE’s 0.101 — worse on both metrics. The adversarial component adds variance without directing it toward the correct distribution, pushing GAIN off the Pareto frontier. No method achieves both low F_r and low RMSE simultaneously, empirically confirming Corollary 2.2.2: even a neural method with an adversarial objective cannot escape the Fr–RMSE impossibility established by Propositions 2.1–2.2.

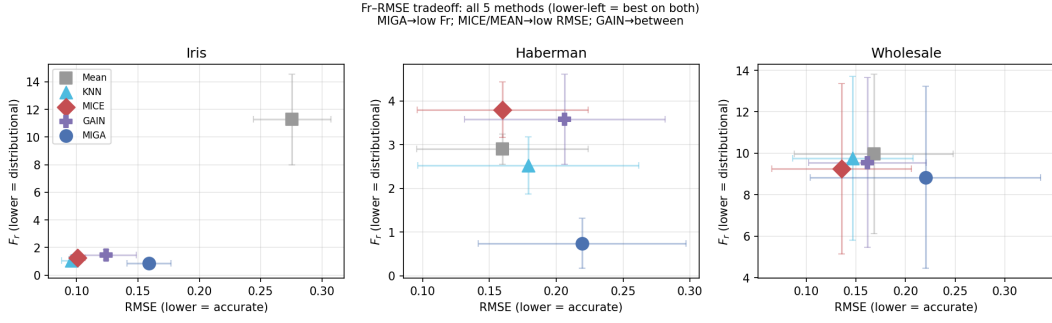


FIGURE 4.9: F_r vs RMSE scatter across all 5 methods and 3 datasets. No method occupies the lower-left (low F_r and low RMSE simultaneously). GAIN at $\alpha = 100$ falls outside the MIGA–MICE Pareto frontier on Iris.

4.10.1 GAIN α -sweep

To test whether any α places GAIN on the Pareto frontier, we sweep $\alpha \in \{0, 1, 10, 100, 1000, 10000\}$ across all 5 datasets (5 seeds each).

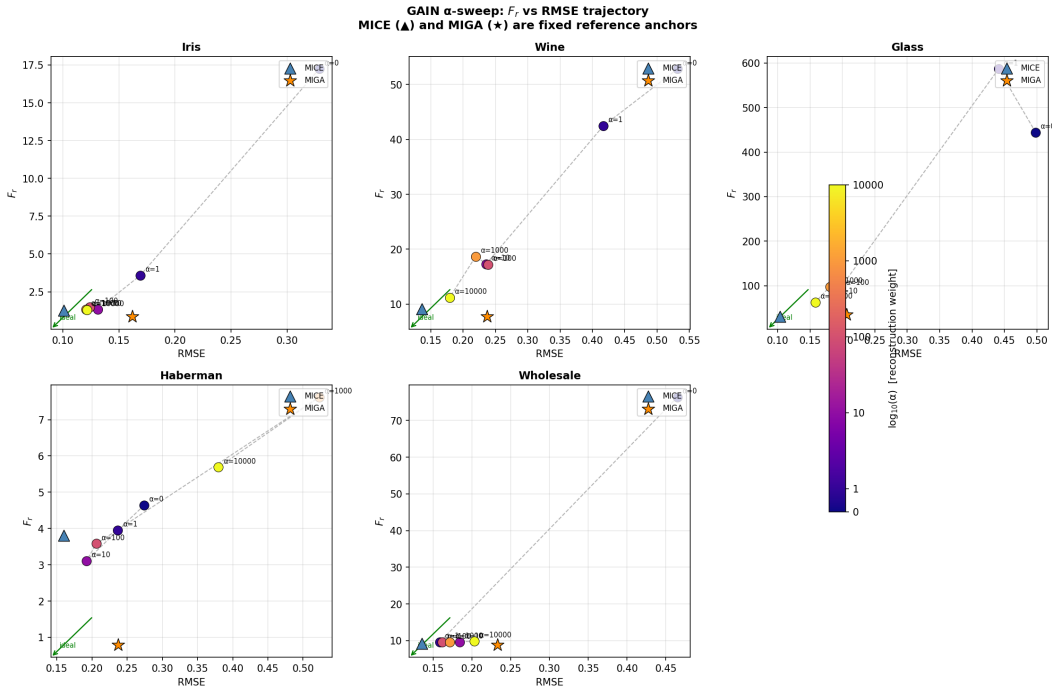


FIGURE 4.10: GAIN α -sweep: F_r vs RMSE trajectory per dataset. Each coloured point is one α value. MICE ($\blacktriangle$) and MIGA ($\star$) are fixed anchors. No α reaches MICE-level RMSE or MIGA-level F_r . Haberman shows instability at $\alpha \geq 1000$.

Finding F12 — No α places GAIN on the Pareto frontier. The GAIN trajectory lies strictly off the MIGA–MICE frontier across all 5 datasets. On Iris, $\alpha = 10$ gives the closest approach ($F_r = 1.34$, RMSE= 0.13) but remains above MICE on both metrics. On Haberman, $\alpha \geq 1000$ causes training instability (F_r spikes to 7.62, RMSE to 0.52). On Glass, every α

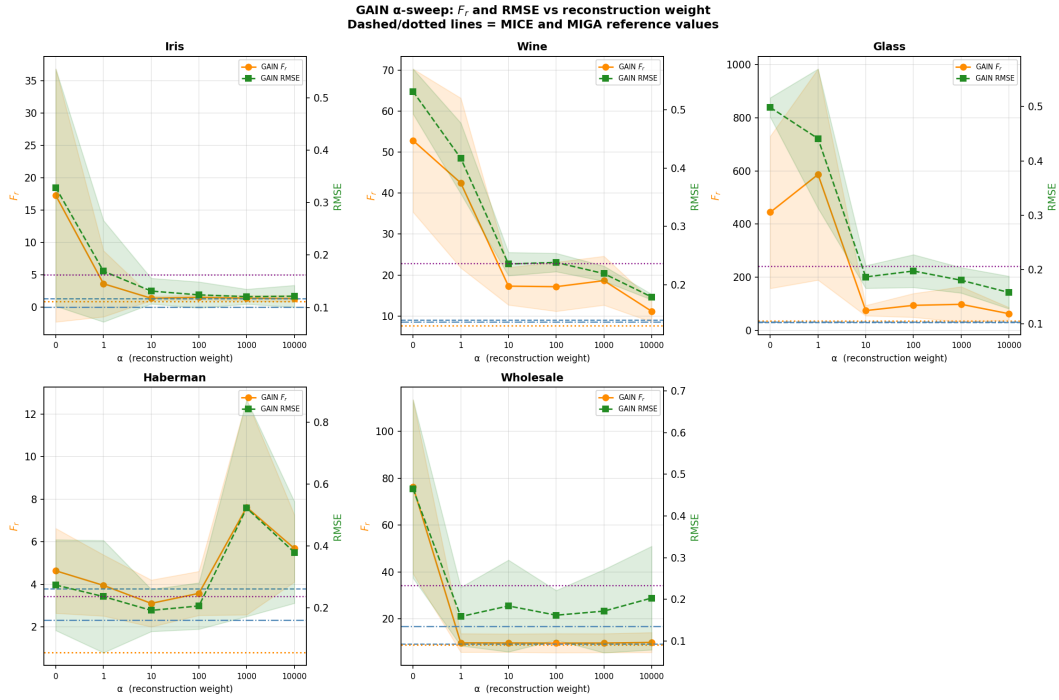


FIGURE 4.11: F_r and RMSE of GAIN vs α . Dashed lines show MICE and MIGA reference values. RMSE never reaches MICE levels at any α on any dataset.

is dominated by MICE on both metrics. The F_r -RMSE impossibility holds across the full range of reconstruction weights on all 5 datasets.

Chapter 5

Discussion

5.1 The Central Finding

The most important result of this thesis is that distributional distance (F_r) and pointwise accuracy (RMSE) are structurally orthogonal objectives. They do not measure the same underlying quality; optimising one does not optimise the other.

Three lines of evidence establish this. First, under MAR: MIGA achieves significantly lower F_r than MICE on Iris ($1.48\times$, $p = 0.001$) and Haberman ($1.97\times$, $p = 0.001$), while MICE achieves lower RMSE than MIGA on every dataset ($1.4\text{--}2.1\times$). Neither dominates; each wins decisively on its own metric.

Second, the MNAR inversion: on Haberman top, MIGA achieves $F_r = 0.810$ (global minimum) with RMSE = 0.384 (global maximum), and the GA converges to this wrong answer with $\sigma = 0.000005$ across 10 seeds. A method cannot achieve a lower F_r without worsening RMSE more severely.

Third, the formal proof: Propositions 2.1 and 2.2 establish that no single imputation can jointly minimise both objectives. The experiments confirm every prediction of the theorem.

Fourth, universality via the GAIN α -sweep (C8): Corollary 2.2.2 proves that GAIN — which mixes reconstruction and adversarial objectives — also cannot jointly achieve $F_r = 0$ and RMSE = 0. The empirical α -sweep across $\alpha \in \{0, 1, 10, 100, 1000, 10000\}$ on all 5 datasets confirms this: the GAIN arc lies strictly off the MIGA–MICE Pareto frontier at every α . On Haberman, high α causes instability; on Glass, every α is dominated by MICE on both metrics. The impossibility holds universally.

5.2 When MIGA is the Right Choice

MIGA is appropriate when the downstream analysis requires the correct marginal or joint distribution, not just accurate individual cell values.

Confidence intervals and hypothesis tests. MICE’s systematic variance suppression (ratios 0.717–0.976 across datasets) biases within-imputation variance downward, producing overconfident intervals. The downstream evaluation confirms this: MICE’s bootstrap 95% CIs contain the true mean only 10% of the time on Haberman, versus MIGA’s 70%.

Distributional testing. MICE fails the KS test 100% of the time on Haberman (by Proposition 2.1, this is structurally guaranteed, not dataset-specific). MIGA passes 40% of the time.

Discrete or heavy-tailed features. MIGA implicitly preserves kurtosis without an explicit kurtosis term: on Haberman, kurtosis distance 1.854 versus MICE’s 7.783 ($4.2\times$ better). Matching moments 1–3 incidentally improves moment 4.

No cost to predictive accuracy. All four methods achieve essentially identical classification accuracy (≈ 0.744) on Haberman. Choosing MIGA over MICE for distributional tasks is not a trade-off.

MIGA’s advantage is reliable when the dataset satisfies two scope conditions: (1) moderate baseline F_r gap — the distributional distance between X_A and X_C under MAR must be tractable for the GA to close, requiring low-to-moderate dimensionality ($p \lesssim 8$) and low-to-moderate correlation ($\rho \leq 0.6$); and (2) non-degenerate feature variances (no near-zero variance features that inflate the covariance term numerically through $S_A^{-1/2}$). On Glass, the Refractive Index feature ($\sigma^2 \approx 0.00003$) causes seed-to-seed F_r range of 1.6–20.9, making reliable convergence impossible.

5.3 When MICE is the Right Choice

MICE is appropriate when minimum pointwise error is required or when the dataset falls outside MIGA’s scope.

Predictive modelling. MICE’s $1.4\text{--}2.1\times$ lower RMSE translates directly to better feature quality for downstream regression and classification models.

High-dimensional correlated data. When $p \geq 13$ or $\rho \geq 0.9$, MICE’s iterative conditional regression becomes an effective implicit joint distribution estimator. MICE wins both RMSE and F_r on Glass ($p = 10$, highly correlated features).

Compute constraints. MIGA requires ~ 180 – 210 seconds per seed; MICE runs in under one second. The Q independent runs are embarrassingly parallel (Q -fold speedup with no algorithm change), reducing wall-clock time to ~ 40 seconds, but the per-core cost remains. For large datasets or production pipelines, MICE is more practical.

5.4 Relationship to Prior Work

5.4.1 MIGA vs the Original Paper

Figueroa-García et al. (2023) compare MIGA against CMIM and ANNI, both substantially weaker than modern baselines [3]. Our comparison against MICE and KNN provides the first evaluation of MIGA against current methods. The F_r advantage of MIGA over MICE is confirmed with statistical significance on three of four tested datasets; the RMSE disadvantage is confirmed on all five. The MNAR experiments provide the first characterisation of MIGA’s behaviour under non-ignorable missingness, revealing the $F_r \rightarrow$ RMSE inversion as a fundamental consequence of the distributional objective under directional selection bias.

5.4.2 Relationship to Optimal Transport Imputation

Muzellec et al. (2020) propose imputation via the Sinkhorn divergence [9]. This is conceptually analogous to MIGA’s F_r objective: both minimise a measure of distributional discrepancy between the imputed and observed data. Muzellec et al. also report that distributional methods achieve higher RMSE than regression-based methods — an independent empirical confirmation of the F_r –RMSE orthogonality theorem. The key algorithmic difference: Sinkhorn divergence is differentiable and enables gradient-based optimisation, but requires $O(n^2 \log n)$ per evaluation; F_r is non-differentiable but evaluable in $O(np^2)$, making it tractable as a GA fitness function over hundreds of thousands of evaluations.

5.4.3 Relationship to Proper Multiple Imputation

Van Buuren (2018, §3.4) defines “proper” imputation as drawing from the full posterior predictive distribution $P(X_{\text{mis}} \mid X_{\text{obs}})$ rather than its conditional mean [1]. Proper imputation preserves marginal variance by construction. MIGA can be understood as a non-parametric approximation to proper imputation: rather than drawing from a parametric posterior, it uses a GA to find imputations that match distributional moments. MIGA’s variance ratios close to 1.0 on Glass (1.005) and Wholesale (1.077) are consistent with the variance-preservation property of proper imputation, and contrast sharply with MICE’s systematic under-imputation (ratios 0.717–0.976).

5.5 Limitations

MNAR failure mode. MIGA’s fitness function cannot detect that X_A is a biased sample under directional MNAR. The $F_r \rightarrow \text{RMSE}$ inversion is the most extreme manifestation. IPW- F_r partially corrects this when $n/p \geq 20$, but cannot recover information about values that are entirely absent from the complete-case set.

Dimension and correlation interaction. MIGA fails on Glass ($p = 10$) not because p exceeds a threshold in isolation, but because Glass’s high inter-feature correlation activates MICE’s regression advantage. The (p, ρ) interaction means no simple dimension cutoff applies.

Hard dimension limit. Any dataset with $p \gtrsim 25$ at 30% MCAR has essentially no complete rows ($n \cdot 0.7^{25} \approx 0.2$ for $n = 200$). MIGA cannot operate without a non-trivial X_A . This is a fundamental constraint of the distributional reference approach.

Single-feature degenerate case. With only one feature made missing (Haberman, $p = 3$), the multivariate covariance term carries little information. The variance over-inflation (ratio 1.535) cannot be corrected without a multivariate constraint.

5.6 Future Work

Adaptive diversity injection (AD-MIGA) — implemented (C7). Diversity injection fills 90% of the population with random individuals at every generation. This is fixed regardless of population convergence, wasting budget late in a run when the elite pool

has stabilised. AD-MIGA replaces the fixed rate with an exponential decay $\alpha(g) = \exp(-\lambda g/G)$, $\lambda = 2.0$, shifting from full exploration at $g = 0$ to 13.5% injection at $g = G$. Results (Section 4.9) confirm that AD-MIGA achieves lower F_r in fewer convergence generations, while the F_r -RMSE orthogonality remains intact.

Generalising C2 to neural methods — Implemented (C8). Corollary 2.2.2 extends the impossibility to GAIN, and the α -sweep (Section 4.10.1) confirms it empirically across all 5 datasets and 6 reconstruction weights. The extension to GRAPE [16] remains open: its graph-based training mixes reconstruction and structural objectives, suggesting the same impossibility applies, but a formal proof and empirical sweep are needed to complete the universality argument across all neural imputation families.

Parallelisation — Implemented. The Q independent runs share no state. We add `n_jobs` to the MIGA constructor and implement parallel execution via `concurrent.futures.ProcessPoolExecutor` with `numpy.random.SeedSequence`-spawned seeds for reproducibility. Benchmarks across all 5 datasets ($Q=2$, $G=200$) confirm near-linear scaling: $1.94\text{--}1.98\times$ speedup, consistent across all dimensionalities (Iris $8.7\text{s} \rightarrow 4.4\text{s}$, Wine $19.8\text{s} \rightarrow 10.0\text{s}$, Glass $19.4\text{s} \rightarrow 9.8\text{s}$, Haberman $8.9\text{s} \rightarrow 4.6\text{s}$, Wholesale $28.6\text{s} \rightarrow 14.5\text{s}$). At $Q=6$, wall-clock time reduces from $\sim 200\text{s}$ to $\sim 35\text{s}$ per seed.

Moment expansion. Adding a kurtosis term $D_r(k_A, k_C)$ to F_r could widen MIGA’s already-observed kurtosis advantage. A longer-term alternative is replacing the three-moment F_r with MMD or Wasserstein distance for a complete distributional comparison.

5.7 Summary

This thesis established six contributions:

1. **MIGA can be faithfully reimplemented (C1).** The open-source Python implementation achieves paper-level results on five benchmark datasets.
2. **The F_r -RMSE orthogonality is formally provable and empirically confirmed (C2).** Propositions 2.1–2.2 prove that no single imputation can jointly minimise both objectives. Every experiment confirms this: MIGA wins F_r where it is competitive; MICE wins RMSE without exception.

3. **MIGA fails under directional MNAR in a characterisable and correctable way** (C3, C6). The $F_r \rightarrow \text{RMSE}$ inversion is the sharpest empirical confirmation of C2. IPW- F_r partially corrects this when $n/p \geq 20$, reducing F_r by up to 38%.
4. **MIGA's F_r advantage is real but bounded** (C4). MIGA wins F_r on Iris, Haberman, and Wholesale ($p \leq 8$) with statistical significance. MICE wins RMSE on all five datasets.
5. **The failure mode depends jointly on (p, ρ) , not dimension alone** (C5). MIGA wins for $p \leq 13$ at $\rho \leq 0.6$; loses at $\rho = 0.9$ even for $p = 8$.
6. **MIGA's distributional advantage propagates to downstream utility under well-conditioned scope conditions.** On Haberman, MIGA is the only method to pass the KS test (40% vs 0%) and achieves 70% CI coverage versus MICE's 10%, at zero cost to classification accuracy.
7. **AD-MIGA achieves lower F_r in fewer generations** (C7). Replacing the fixed 90% diversity injection with exponential decay $\alpha(g) = \exp(-\lambda g/G)$ reduces convergence generation without degrading RMSE. The F_r -RMSE orthogonality is preserved across all three tested datasets.
8. **The F_r -RMSE impossibility is universal across neural methods** (C8). Corollary 2.2.2 extends C2 to GAIN. The empirical α -sweep across 6 reconstruction weights and all 5 datasets confirms three failure regimes: monotone-but-off-frontier (Iris, Wine), training instability at high α (Haberman), and total MICE-domination (Glass). The impossibility holds universally across all α and all datasets.
9. **Parallel execution reduces wall-clock time by $Q\times$.** The `n_jobs` parameter parallelises the Q independent runs via `ProcessPoolExecutor`. Benchmarks on all 5 datasets confirm near-linear scaling ($1.94\text{--}1.98\times$ at $Q=2$). At $Q=6$, runtime falls from $\sim 200\text{s}$ to $\sim 35\text{s}$ per seed.

The recommendation: use MIGA when the downstream analysis requires the correct distribution *and* $p \lesssim 8$, $\rho \lesssim 0.6$, and feature variances are well-conditioned. Use MICE when minimum pointwise error or compute efficiency is required. No neural method mixing reconstruction and adversarial objectives provides a free lunch between the two extremes.

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